Haixiang Sun

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Education

Purdue University West Lafayette, United States

Ph.D in Industrial Engineering *Aug. 2024 - Present*

- Research Direction: Machine learning and operations research, robust decision making under uncertainties

ShanghaiTech Univerisity Shanghai, China

Master in Computer Science *Sept. 2021 - Jul. 2024*

- Research Direction: Machine learning theories, interpretability of neural networks
- Courses: Convex Optimization, Machine Learning, Algorithm Design and Analysis
- (major) GPA: 3.86/4.0, (overall) GPA: 3.84/4.0

Beijing Forestry University Beijing, China

Bachelor's degree in Information Management and Information System *Sept. 2017 - Jun. 2021*

- GPA: 3.83/4.0, Outstanding bachelor graduate
- Thesis title: Structural causal network learning and application of rapid transit in a certain city (*Excellent Graduation Thesis, Top 1%*)

Research interests

I have strong interest in topics related to practical application of machine learning. My research mainly focus on the theory of optimization, and discovering the relationship between machine learning algorithms and operations research scenarios. I am particularly intrigued by the theoretical research in machine learning, including implicit models, optimization under uncertainty, and especially their applications to valuable data analysis processes, such as causal discovery, financial data analysis, energy scheduling, policy making, and among other real applications.

Research Experience

Optimization and robust decision making under uncertainties *Aug. 2024 - Present*

Current research direction during Ph.D. studies

- Integrate optimization insights into end-to-end decision-making pipelines, ensuring that the structure and properties of optimization problems are effectively leveraged within the learning process.
- Employ diffusion models for scenario generation under varying renewable energy profiles, with optimization-guided training to enable more reliable scheduling decisions under uncertainty.
- Leverage optimal transport and distributionally robust optimization to generalize outliers, enabling robust modeling of rare or unforeseen events such as accidents or extreme system shocks. This facilitates realistic simulations that support proactive risk management and loss mitigation.

Differentiable optimization and implicit neural networks *Sept. 2021 - Jun. 2024*

During the master period at ShanghaiTech University

- Established the optimization layer in neural networks: explored the integration of convex optimization problems with neural networks as implicit layers, enabling the forward and backward propagation akin to explicit layers.
- Designed an ADMM-based algorithm to enhance the scalability of large-scale optimization tasks, enabling faster and more reliable solutions for energy scheduling and optimal power flow problems.

- Explored various computational methods for implicit differentiation through the application of the Implicit Function Theorem, and considered the impact and potential enhancements of inexact gradients during the network training process.
- Investigated the theory and current challenges within a typical implicit model (Deep Equilibrium Model), particularly in the context of initialization and implicit bias stemming from the neural tangent kernel. Additionally, elucidated its performance through the lens of neural collapse.

Scalable methods for causal discovery *Oct. 2020 - Present*
Long-term ongoing research interest

- Investigated various methods of causal discovery, especially for large-scale graphs, applying divide-and-conquer methods and identification of Markov Blanket to improve the efficiency of causal discovery.
- Proposed a novel likelihood function that incorporates specific causal parameters into the kernel of a Gaussian process and obtained the corresponding causal graph by optimizing the likelihood with DAG constraints, considering design a new kernel to represent the causal relationship.
- Employed Gaussian process state space models and system identification methods to account for the potential influence of external latent confounders on real causal relationships, such as rapid transit data, thus improving the accuracy of causal discovery.
- Discovered the link between causal discovery and score matching, and tried tower decomposition for topological ordering to further promote the performance.

Publications

◦ **Talk Less, Work More: Communication-Efficient Decentralized Stochastic Approximation**
NeurIPS 2026, Atlanta, GA
T. Cao, **H. Sun**, Y. Xu

◦ **Beyond Empirical Support: Structured Outlier Generation via Sinkhorn Optimal Transport**
NeurIPS 2026, Atlanta, GA
H. Sun, A. L. Liu

◦ **Towards Reliable LLM Evaluation: Correcting the Winner's Curse in Adaptive Benchmarking**
NeurIPS 2026, Atlanta, GA
Y. Xu, J. Zhang, **H. Sun**, Z. Zhou, T. Cao, V. Aggarwal

◦ **Hierarchical Causal Structure Discovery from Layered Topological Orderings**
Accepted by *International Journal of Data Science and Analytics*
H. Sun, P. Tian, Z. Zhou, A. L. Liu

◦ **Robust Generalization with Adaptive Optimal Transport Priors for Decision-Focused Learning**
AISTATS 2026, Tangier, Morocco
H. Sun, A. L. Liu

◦ **Understanding Representation of Deep Equilibrium Models from Neural Collapse Perspective**
NeurIPS 2024, Vancouver, BC
H. Sun, Y. Shi

◦ **Alternating Differentiation for Optimization Layers**
ICLR 2023, Kigali, Rwanda
H. Sun, Y. Shi, J. Wang, H. D. Tuan, H. V. Poor, D. Tao

- **Application of Calculus in Determinants**
College Mathematics, 2019, 35(04): 122-126
H. Sun, Z. Niu

Papers Under Review

- **CARVE: Breaking Data Barriers in Chip Placement by Harnessing Reusable Expertise**
Under review
J. Zhang, **H. Sun**, Y. Xu, V. Aggarwal, Z. Wan

- **Reliable Replay through Spatial Coherence in Online Continual Learning**
Under review
H. Sun, J. Zhang, Y. He, Y. Xu, V. Aggarwal, B. Bhargava, A. L. Liu

- **Suppressing Pressure, Amplifying Evidence: Self-Guided Attention Steering to Mitigate Sycophancy and Stubbornness**
Under review
Y. He, M. Xu, **H. Sun**, D. Li, Y. Wang, L. Wang, K. Chen, C. Li, C. Liu, B. Bhargava, C. Gao

- **Diff2SP: Diffusion Models for Correlated Scenario Generation in Stochastic Programming**
Major Revision under *INFORMS Journal on Data Science*
H. Sun, A. L. Liu

- **Efficient Causal Structure Learning via Modular Subgraph Integration**
Under review
H. Sun, P. Tian, Z. Zhou, J. Zhang, P. Li, A. L. Liu

- **Core-Halo Decomposition: Decentralizing Large-Scale Fixed-Point Problems**
Under review
T. Cao, **H. Sun**, Y. Xu, J. Zhang, X. Wu, Z. Zhou, J. He, J. Chen

Talks

AAAI Bridge program - Combining AI and OR/MS for Better Trustworthy Decision Making, Feb 2025, Philadelphia, PA

INFORMS Annual Meeting, Oct 2025, Atlanta, GA

Technical skills

Programming Languages: Python, MATLAB, LaTeX, C/C++, Julia (currently learning)

Language: Chinese, English

Honors & Awards

Theodore and Isabel Williams Fellowship in Industrial Engineering

Outstanding bachelor graduate in Beijing

Excellent Graduation Thesis in Beijing

First prize of Olympic mathematical contest

Teaching Assistant

ShanghaiTech University	SI152 - Numerical Optimization	Spring 2022
	Create linear programming assignments, grade homework and projects, and provide support through office hours.	
	SI251 - Convex Optimization	Fall 2022
	Answer questions on fundamental concepts in convex optimization, grade homework, and evaluate exams.	
Purdue University	IE332 - Computing in Industrial Engineering	Fall 2024, Spring 2025
	Conduct lab sessions and teach topics such as MySQL, database management, and website development.	
	IE535 - Linear Programming	Fall 2025
	Answer questions about linear programming and simplex methods, grade homework, and evaluate exams.	
	IE546 - Economic Decisions in Engineering	Spring 2026

Reviewer

NeurIPS (2024, 2025, 2026), AAAI (2025, 2026, 2027), ICLR (2025, 2026, 2027), AISTATS (2025, 2026, 2027), ICML (2025, 2026)